

Comprehensive Review and Phenomenological Analysis: Random Sequential Adsorption of Worldlines in Toroidal Cosmology

Introduction to the Path-Centric Cosmological Paradigm

The manuscript under review, which elucidates a packing model of a closed universe built from the random sequential adsorption (RSA) of unit-rapidity worldlines on a comoving torus, represents a profound and highly original theoretical departure from traditional field-theoretic

cosmological frameworks.¹ The prevailing models of modern cosmology—most notably the Λ CDM paradigm and inflationary theory—presuppose the existence of a continuous spacetime manifold upon which various quantum and classical fields fluctuate, interact, and evolve. In these standard models, matter and energy are treated as localized excitations of these underlying continuous fields.

Conversely, the model presented by Bentley and Bentley operates on a strictly discrete, path-centric, and purely kinematic foundation. The universe is constructed from nothing but mutually excluding, closed trajectories that evolve over a conformal time loop.¹ By enforcing an absolute, everywhere-in-time exclusion rule—dictating that no two worldlines may intersect at any point in their entire history—the model shifts the genesis of spatial and thermodynamic structure away from local differential equations and physical potentials, rooting it instead in global, topological combinatorics.¹ "Matter," within this framework, is simply defined as the surviving population of trajectories that successfully navigate the temporal loop without collision.

This report provides an exhaustive, multi-disciplinary review of the manuscript's core mathematical architecture, empirical findings, and computational methodologies. The analysis comprehensively contextualizes the model within the established literature of irreversible deposition kinetics, topological cosmology, and the emerging physics of complex attractor-driven dynamics. Finally, the report delineates critical unexplored theoretical territories and prescribes a suite of high-resolution numerical experiments and algorithmic enhancements necessary to elevate this exploratory framework into a mature, rigorously verifiable cosmological analogue.

Theoretical Architecture and Mathematical Framework

The Sinusoidal Expansion and the Unit-Rapidity Constraint

The fundamental entity within this universe is the individual worldline, which is parameterized over a finite conformal time interval defined as $z \in (0, \pi)$.¹ The universe itself is governed by a simple cyclic scale factor $a(z) = \sin z$, dictating an expansion from a "big bang" singularity at $z = 0$, reaching maximum volumetric expansion at the turnaround $z = \pi/2$, and symmetrically contracting to a "big crunch" at $z = \pi$.¹ Along each spatial axis, the trajectory of a worldline is mathematically expressed as a superposition of integer-frequency sine terms:

$$x(z) = a_1 \sin z + \sum_{k \geq 2} a_k [\sin(kz + f_k) - \sin f_k] \quad .^1$$

This specific formulation is highly intentional. The subtraction of the constant $\sin f_k$ enforces a strict topological closure rule: regardless of the chosen phase f_k , the spatial coordinate $x(z)$ is rigorously pinned to zero at both $z = 0$ and $z = \pi$. Consequently, every generated path must begin exactly at the bang and terminate exactly at the crunch, ensuring that all worldlines share the same ultimate temporal boundaries.¹

The most critical and innovative structural imposition on these trajectories is the unit-rapidity constraint, mathematically defined as $\sum_{k \geq 2} |ka_k| = 1$.¹ In the context of standard relativity, the speed of light c serves as an inviolable local property of the spacetime metric. In the Bentley model, however, the speed limit is intrinsically wired into the parametric definition of the particle's history as a unitary kinematic budget.¹ Because the derivative of the spatial position with respect to conformal time, $|dx/dz|$, is bounded by the sum of its frequency components, the rapidity constraint guarantees that no worldline can exceed a maximum velocity of 1. The amplitudes of all non-trivial frequency components are thus fundamentally interdependent; kinematic budget expended on high-frequency, rapid oscillations inherently depletes the available budget for lower-frequency, macroscopic movements.¹

The Decoupling of Comoving Coordinates and Exact Homogeneity

While the dynamic oscillatory terms ($k \geq 2$) are strictly bound by the rapidity constraint, the fundamental frequency term ($k = 1$) is explicitly exempted.¹ The amplitude a_1 is drawn freely from a uniform distribution $a_1 \in (-1, 1)$. This term governs the static comoving position of the worldline, functioning analogously to the Hubble flow in standard Friedmann-Lemaître-Robertson-Walker (FLRW) cosmology. In comoving coordinates, defined

as $X = x / \sin z$, the fundamental term isolates the static placement of the trajectory from its dynamic wiggles.¹

By mathematically divorcing the static comoving position from the dynamic rapidity budget, the model guarantees spatial homogeneity as an intrinsic property of the probability measure. The spatial placement of a worldline costs zero kinetic energy. Consequently, the initial uniform draw of a_1 generates a perfectly homogeneous spatial distribution. Because the subsequent dynamical acceptance or rejection of a worldline relies purely on relative geometric exclusion rather than absolute spatial placement, the model eradicates any notion of a preferred origin or "center" of the universe.¹

Toroidal Topology and the Minimum Image Convention

To computationally and theoretically support a center-less universe without introducing pathological edge effects, the model assumes that the comoving spatial volume is a

d -dimensional torus ($d = 2$ or $d = 3$).¹ The physical bounds of the universe along any given spatial axis are defined by the expanding and contracting envelope $\pm \sin z$.

In the toroidal geometry, a worldline that crosses the boundary at $+\sin z$ does not reflect or terminate; rather, it instantaneously re-enters the spatial volume through the opposite face at $-\sin z$, continuing its trajectory with unbroken momentum.¹ In the normalized comoving coordinate system, this equates to the statement that the comoving coordinate X exists on a periodic circle of circumference 2, where $X = +1$ and $X = -1$ map to the exact same physical location.¹

The choice of a 3-torus topology is highly relevant to contemporary debates in theoretical cosmology. The global topology of the observable universe remains an open empirical question, with cosmic microwave background (CMB) measurements from WMAP and Planck unable to definitively rule out multiply connected spaces like the 3-torus, which possess everywhere-zero curvature but finite extent.²

However, recent theoretical investigations into the quantum origins of the universe have cast doubt on the viability of toroidal topologies in semiclassical regimes. For instance, Guth and Vilenkin's analysis of the quantum creation of a toroidal universe from "nothing" utilizing the Wheeler-DeWitt equation and Casimir energies reveals that the corresponding instantons are fundamentally singular.⁶ They argue that a 3-torus spacetime is geodesically incomplete and its quantum creation cannot be reliably described without invoking unknown Planck-scale physics.⁶

The Bentley model elegantly bypasses these semiclassical quantum gravity pathologies. Because the universe is constructed kinematically from discrete, classical paths rather than emerging from a continuous quantum vacuum fluctuation, it is entirely divorced from the Wheeler-DeWitt equation and the associated instanton singularities.¹ The torus in this

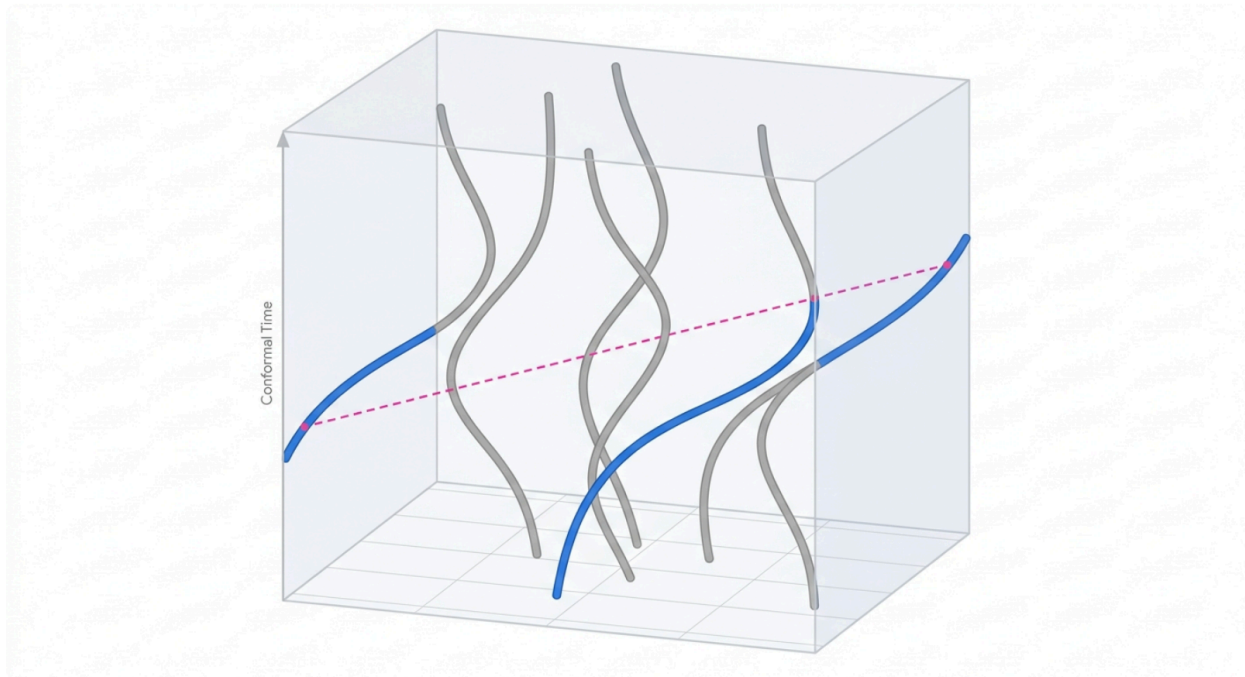
framework serves strictly as a periodic container for kinematic exclusion, shielding the findings from the topological instabilities identified in recent quantum cosmology literature.

To compute valid distances and evaluate path intersections within this periodic space, the

model correctly utilizes the minimum image convention.¹ For any two points i and j , the distance is calculated not across the naive Cartesian grid, but across the shortest possible path taking into account the periodic boundaries, effectively computing

$$\min(|X_i - X_j|, L - |X_i - X_j|)$$

Worldline Topology and Minimum Image Boundary Crossings



Conceptual representation of sinusoidal worldlines evolving through conformal time on an unwrapped comoving torus. Paths exiting one face of the spatial bounding volume instantaneously re-enter the opposite face. Distances to determine intersections are computed via the shortest periodic path (minimum image).

Random Sequential Adsorption (RSA) on a Spacetime Grid

The population of this universe is executed via Random Sequential Adsorption (RSA). RSA is a classical non-equilibrium statistical mechanics process characterized by the irreversible, random placement of mutually excluding objects.¹ In traditional RSA paradigms, objects—such as hard spheres, discs, or polymers—are deposited into an empty space. If a newly proposed object overlaps with any previously accepted object, it is permanently rejected. This process continues until a saturation point, or "jamming limit," is reached, beyond which no further

objects can be accommodated.¹³

The Bentley model introduces a profound dimensional leap to standard RSA. The objects being packed are not static geometric volumes, but fully extended 1D curves embedded in a

$(d + 1)$ -dimensional spacetime.¹ The exclusion rule operates globally across the entire temporal loop: a candidate worldline must avoid every previously accepted worldline at every single moment in conformal time simultaneously.¹

To evaluate this computationally, the continuous time loop is sampled at T discrete, evenly spaced intervals.¹ At each interval, every path is assigned an exclusion radius of $1/T$ in comoving coordinates. Therefore, two paths intersect if their comoving separation falls below $2/T$ per axis at any evaluated timestep.¹ Because the exclusion radius strictly shrinks as the

temporal resolution T increases, the question of "how many worldlines fit" becomes a mathematically rigorous scaling problem.

The algorithmic implementation of this collision detection is highly optimized. By utilizing comoving coordinates, the exclusion threshold collapses into a constant $2/T$ across all timesteps, neutralizing the effects of the physical scale factor $a(z)$ during collision checks.¹

This constancy allows the authors to deploy a uniform spatial hash grid, achieving an $O(NT)$

collision test complexity rather than the prohibitive $O(N^2T)$ scaling that naive all-pairs distance checking would require.¹ This optimization enables deployment on consumer GPU

hardware, achieving resolutions up to $T = 200$ in 3+1 dimensions and $T = 400$ in 2+1 dimensions.

Empirical Findings: Dynamical Capacity, Geometry, and Scaling

The Jam-Count Exponent and Sub-Dimensional Capacity

The most quantitatively striking result of the manuscript is the pristine power-law scaling of the jammed worldline count against the temporal resolution: $N_{sat} \sim T^D$.¹ Across the fully

resolved ladder, the data yields highly stable scaling exponents of $D_{3+1} \approx 2.33$ and $D_{2+1} \approx 1.38$.¹

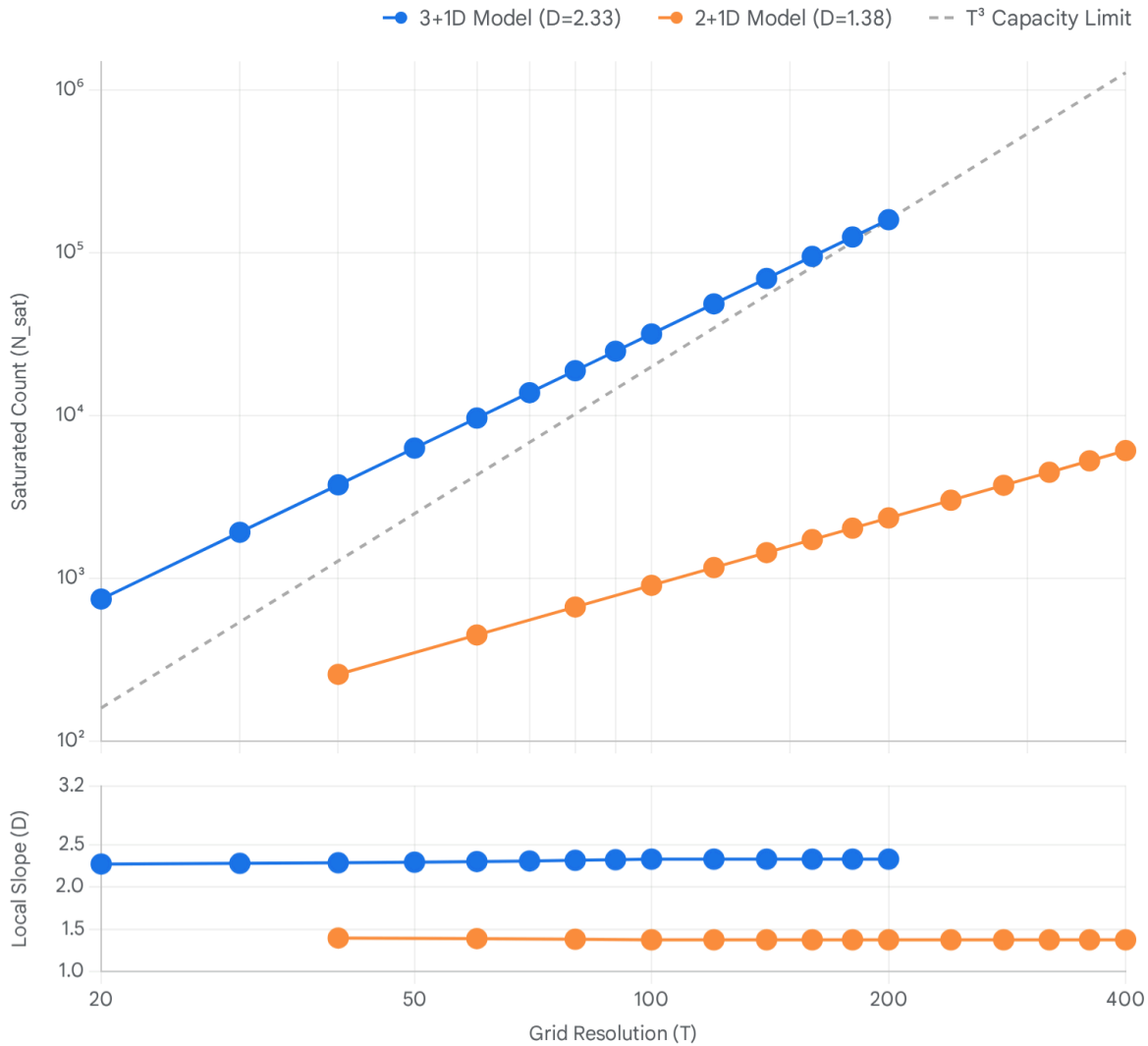
This scaling behavior reveals a profound physical reality regarding the model's capacity. In a scenario governed by pure spatial packing without temporal constraints, the number of

worldlines would scale directly with the volume of the comoving grid, yielding $N_{sat} \sim T^d$.¹

The fact that the measured exponent D sits strictly below the spatial dimension d (e.g., 2.33 versus 3.0) represents the severe combinatoric penalty imposed by the global temporal constraint.

To survive the RSA process, a candidate worldline must successfully thread the microscopic, shifting gaps between all existing paths across T simultaneous temporal slices. The number of trajectories capable of navigating this dynamic labyrinth grows significantly slower than the raw volume available in any single static slice.¹ The authors correctly classify D as a "packing-number exponent," representing a dynamic capacity rather than a static geometric feature.

Jam-Count Scaling Reveals Sub-Dimensional Universe Capacity



Top panel: Saturated worldline count (N_{sat}) as a function of grid resolution (T) on a log-log scale. The 3+1 dimensional model scales at $T^{2.33}$, significantly below the theoretical spatial limit of T^3 . Bottom panel: The local pairwise slope remains flat across the resolution ladder, confirming the absence of crossovers and validating the power-law exponent.

Data sources: [Bentley & Bentley \(2026\)](#)

Absence of Geometric Fractality

Given a fractional exponent like 2.33, the immediate reflex in statistical physics is to hunt for an underlying static fractal structure—a set whose correlation or box-counting dimension

matches the exponent. The authors systematically dismantle this hypothesis.¹ By pooling all timesteps of a saturated jam into a single composite point cloud, the authors measured the local box-counting dimension across varying length scales. The analysis reveals a dynamic sweep rather than a stable fractal plateau: at the finest resolved scales, the local dimension is ≈ 1.3 , reflecting the smooth 1D nature of individual worldline curves; at coarse scales, the dimension rises to ≈ 2.9 , confirming that the bundle is ultimately space-filling.¹ The dimension merely crosses 2.33 in passing. This effectively uncouples the dynamic packing exponent D from any static geometric shadow, underscoring the unique nature of space-time RSA constraints compared to traditional spatial RSA.¹

Exact Homogeneity and the Grassberger-Procaccia Algorithm

Despite the severe combinatoric culling imposed by the RSA process, the surviving matter distribution remains exactly spatially uniform at every evaluated timestep. The root mean square (rms) comoving spread remains perfectly flat at $\sqrt{1/3} \approx 0.5774$ over the entire conformal loop—the exact theoretical value for a uniform distribution on a torus.¹ To further validate this homogeneity, the authors utilized the Grassberger-Procaccia algorithm to compute the correlation dimension D_2 of the equal-time slices.¹ The algorithm analyzes the spatial correlation integral $C(r)$, which counts the number of particle pairs separated by a distance less than r , seeking the scaling relation $C(r) \sim r^{D_2}$.¹⁰ Crucially, the implementation of this algorithm necessitates rigorous adherence to the periodic boundary conditions. The authors correctly utilized a wrapped periodic k-d tree to compute the minimum image distance between pairs.¹ Under this strict periodic metric, both spherical and cubic probes converged onto the exact space-filling values ($D_2 = 3.01$ for 3D and 2.03 for 2D).¹ The authors astutely demonstrate that failure to wrap distances in the Grassberger-Procaccia estimator—treating the periodic data as if it possessed open boundaries—results in artificial edge effects that depress the correlation dimension to a spurious ≈ 2.89 .¹ This control test unequivocally confirms that the uniform comoving measure, initialized by the free a_1 term, survives the extreme pruning of the RSA jam flawlessly.

Kinematics, Parity Selection, and Thermodynamics

Broken Symmetry and Odd-Frequency Dominance

A deep structural asymmetry exists within the model regarding the parity of the frequency

components k . Because all worldlines must close to exactly zero at the bang ($z = 0$) and crunch ($z = \pi$), the allowable phase freedoms differ depending on whether the frequency is odd or even.¹

For odd frequencies, a simple sign flip of the amplitude equates exactly to a phase shift by π . Consequently, odd components naturally satisfy the topological closure offsets and require no explicit continuous phase modulation.¹ In stark contrast, even frequencies demand an explicit continuous phase parameter $f_k \in [0, \pi)$ to properly cancel the re-pinning constants.¹ This mathematical asymmetry manifests as a profound physical symmetry breaking under the pressure of the RSA process. The spatial profile of an odd frequency b dictates that the worldline sits at a velocity extremum—meaning it is exactly at rest—at the turnaround ($z = \pi/2$). Conversely, even frequencies swing through the turnaround at maximum speed.¹ The empirical data demonstrates a massive dynamical selection in favor of odd frequencies. In the sparse, early stages of packing, even frequencies are accepted at rates of 20-27%. However, in the final, highly crowded quartile preceding the jam, even frequency acceptance plummets to 3-13%.¹ Ultimately, 82.7% of all surviving frequencies in the jammed state are odd.¹ The physical intuition is clear: in an exceedingly dense universe, trajectories that remain stationary at the moment of maximum spatial expansion possess a vastly superior probability of dodging collisions. The geometric exclusion inherently selects for "cold," stationary matter over mobile trajectories.

The Attenuated Arcsine Law and Phase Selection

For the minority of worldlines that do possess an even frequency (the "mobile component"), the explicit phase freedom allows for a closed-form kinematic prediction. At the turnaround, the peculiar velocity reduces simply to $v = |\cos f|$.¹ Assuming a uniform distribution of the phase parameter f , the resulting speed of a mover follows a parameter-free arcsine distribution: $P(v) = 2/(\pi\sqrt{1-v^2})$.¹

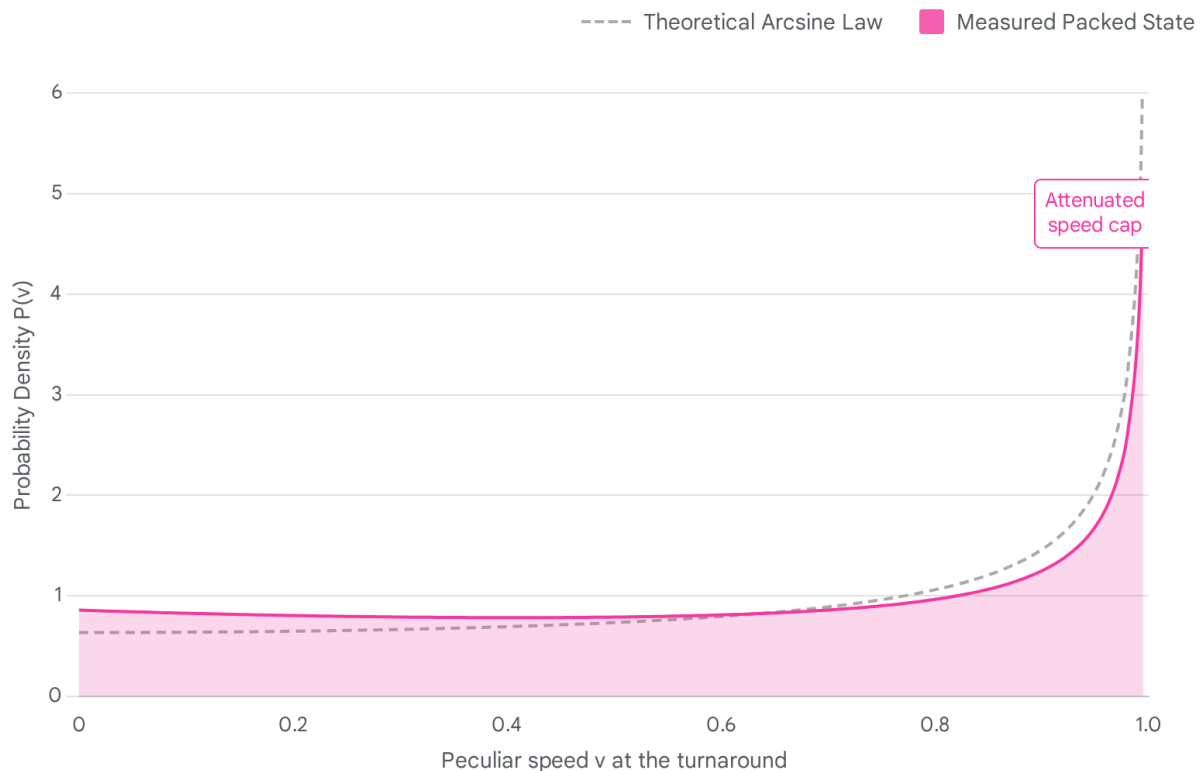
This theoretical distribution is physically striking, featuring an integrable divergence, or pile-up, exactly at the speed cap $v = 1$, mirroring relativistic bounds without requiring an explicit Lorentz transformation.¹ However, the RSA packing process deeply modifies this pristine kinematic law.

The measured speed distribution of the jammed state qualitatively retains the arcsine shape but introduces a severe, measurable tilt toward slow movers.¹ The empirical mean squared velocity $\langle v^2 \rangle$ drops from the theoretically predicted 0.500 down to 0.404.¹ Because the RSA acceptance rules contain no explicit reference to velocity, this attenuation proves that the

jamming process aggressively selects on phase.¹

Trajectories with phases near 0 or π traverse vast amounts of comoving ground at maximum speed precisely when the universe is at maximum expansion. Such high-sweeping trajectories are overwhelmingly likely to intersect existing paths and are thus mercilessly rejected. Consequently, the surviving mobile population is heavily biased toward phases near quadrature, effectively cooling the kinematic profile of the universe purely through geometric filtering.

Jamming Modifies Kinematics: The Attenuated Arcsine Law



Probability density of peculiar speeds at the turnaround. The theoretical proposal (gray line) follows an arcsine law with a relativistic pile-up at the speed cap ($v=1$). The measured packed state (magenta area) retains the overall shape but exhibits a marked tilt toward slower movers, reflecting the rejection of high-sweeping volume trajectories during random sequential adsorption.

Data source: [A packing model of a closed universe \(Bentley & Bentley, 2026\)](#).

Emergence of a Matter-Like Equation of State

Thermodynamically, the universe can be characterized by its equation of state parameter $w = P/\rho$, where P is kinetic pressure and ρ is energy density. Utilizing the kinetic pressure derived from the surviving mobile component, $w = \frac{1}{3} \langle E v^2 \rangle / \langle E \rangle$, the manuscript calculates the equation of state at the turnaround.¹

The resulting measurement yields $w = 0.070$, a value that remains incredibly stable across the entire evaluated resolution ladder from $T = 20$ to $T = 200$.¹ This value is distinctly matter-like ($w \ll 1/3$), indicating a "dust" dominated universe. Remarkably, this outcome is highly robust, holding identical to within 1% under two completely independent mass dictionaries: one treating energy as proportional to frequency ($E \propto \sum b_k$, akin to waves) and another treating energy as proportional to the spatial arc length ($E \propto \int ds$, akin to strings).¹

The unselected, raw arcsine kinematic proposal would yield an equation of state of $w \approx 0.087$. The measured $w = 0.070$ sits nearly 20% below this baseline, which precisely quantifies the thermodynamic impact of the phase-selection tilt.¹ A universe packed entirely with unit-rapidity trajectories inherently cools itself into a cold, matter-dominated state simply through the combinatorics of spatial exclusion.

Comprehensive Contextualization with Related Literature

To fully evaluate the viability and novelty of the Bentley model, it must be rigorously benchmarked against established theoretical paradigms. The model synthesizes concepts from three distinct domains of physics: the kinetics of irreversible deposition, the topology of early-universe string theory, and the nonlinear dynamics of complex active matter.

1. Kinetics of RSA for Extended and Anisotropic Objects

The RSA of complex geometries has a rich history in statistical mechanics, originating from Paul Flory's 1939 models of polymer chains and Alfréd Rényi's 1D car-parking problem.¹³ While the packing of simple isotropic spheres is well understood, the RSA of anisotropic, extended objects—such as k -mers, rectangles, spherocylinders, and line segments (needles)—exhibits drastically different asymptotic scaling properties.²⁰

Property	Standard RSA (Isotropic Spheres/Discs)	Anisotropic RSA (Ellipses/Polymers /Needles)	Bentley Model (Space-Time Worldlines)

Exclusion Space	Static 2D/3D Euclidean grid.	Static 2D/3D continuum or discrete lattice.	Dynamic 4D Conformal Spacetime loop.
Approximation to Jamming	Feder's Law: $\theta(t) \sim \theta(\infty) - ct$	Feder's Law modified by degrees of freedom: t^{-1/d_f} (where $d_f > d$).	Scales dynamically by temporal resolution: $N_{sat} \sim T^D$.
Jamming Limit	Finite, precisely defined density (e.g., ≈ 0.547 for 2D circles).	Dependent on aspect ratio; infinitely thin needles exhibit no finite geometric jamming limit without constraints.	Finite dynamical capacity limited by combinatorics of temporal non-intersection ($D < d$).

Table 1: Comparative analysis of packing mechanics between classical RSA and the Bentley space-time model. ¹

In classical extended RSA, the asymptotic approach to the jamming limit $\theta(\infty)$ is governed by Feder's law, heavily influenced by the object's degrees of freedom (d_f). For an infinitely thin line segment (a needle), the proper area is zero, meaning an infinite number of needles can continually be packed without ever reaching a true geometric jamming saturation.²⁰ The worldlines in the Bentley model are mathematically akin to these 1D needles, tracing paths through a 4D space. However, they evade the infinite-capacity paradox because their exclusion radius ($2/T$) shrinks explicitly as a function of the resolution scale T .¹ Instead of tracking surface coverage $\theta(t)$ over continuous deposition time, the Bentley model tracks the total carrying capacity N_{sat} against the inverse grid resolution, yielding the unique sub-dimensional $T^{2.33}$ scaling.¹ This places the model in an entirely novel class of packing problems, bridging static geometry with dynamic, resolution-dependent intersection theory.

2. Topological Echoes of String Gas Cosmology (SGC)

The architectural premise of a toroidal universe populated by looping, 1D extended objects strikes profound parallels with String Gas Cosmology (SGC), initially formulated by Robert

Brandenberger and Cumrun Vafa.²⁴ SGC is a string-theory-based alternative to standard inflationary cosmology designed to resolve the initial singularity by relying on the unique thermal and winding properties of a gas of closed strings within a compact topology.²⁷

Cosmological Framework	Fundamental Entity	Topological Constraints	Thermodynamic State at Extremum	Mechanism of Structure Generation
String Gas Cosmology	Closed Strings (Momentum, Oscillatory, and Winding modes).	Compact Torus T^d , governed by T-duality ($R \leftrightarrow 1/R$).	Hagedorn Phase (Quasi-static, pressureless dust $w \approx 0$).	Thermal fluctuations of the string gas.
Bentley Packing Model	Unit-Rapidity Worldlines (Sinusoidal frequency components).	Comoving Torus T^d , evaluated over a conformal time loop $(0, \pi)$.	Matter-like state ($w = 0.070$), heavily favoring odd (stationary) frequencies.	Pure geometric exclusion (Random Sequential Adsorption).

Table 2: Conceptual alignment between String Gas Cosmology and the Bentley path-packing universe.¹

In SGC, strings exhibit momentum modes (representing center-of-mass motion) and winding modes (counting the number of times a string wraps around the periodic dimensions of the torus). The winding modes actively resist the expansion of the universe, maintaining it in a quasi-static, high-temperature "Hagedorn phase" dominated by a pressureless dust equation of state.²⁵

The Bentley worldlines function as a classical analogue to these string states. The fundamental comoving coordinate (a_1) maps to the momentum modes, while the rapidly oscillating, periodic high-frequency components effectively "wind" through the temporal loop.¹

Remarkably, the Bentley model’s measured turnaround equation of state ($w = 0.070$) independently arrives at the same pressureless, matter-dominated thermodynamic regime as SGC’s Hagedorn phase.¹ The convergence of these distinct frameworks—one driven by quantum string thermodynamics and the other by purely geometric classical exclusion—suggests that topological constraints on extended 1D structures naturally engender

matter-dominated cosmologies.

3. Parallels to Attractor-Driven Active Matter

The authors themselves pose the question of whether the jammed RSA state can be formally classified as a "strange attractor".¹ This hypothesis aligns neatly with recent advancements in the physics of active matter, particularly the concept of "attractor-driven matter" pioneered by Valani and Paganin.²⁹

In traditional active matter systems (e.g., flocking birds or bacterial suspensions), particles interact via explicit aligning forces or purely repulsive physical potentials, leading to clustering, jamming, or macroscopic order.³¹ In attractor-driven matter, individual particles are endowed with internal, chaotic state spaces—often modeled by nonlinear dynamical equations like the Lorenz system—and the particle's macroscopic physical trajectory is directly driven by the evolution of this internal strange attractor.²⁹

In the Bentley model, the individual worldlines do not possess internal differential equations. However, the RSA algorithm itself functions as an implicit, global basin of attraction.¹ The

uniform initial parameter space of frequencies and phases (b, f, a_1) is brutally filtered by the global non-intersection constraint. As the packing iterates, the surviving population is inexorably drawn onto a highly specific, measure-zero-ish manifold characterized by odd-frequency dominance, coprime tuples, and quadrature phases.¹ The mechanism is functionally identical to attractor-driven matter: a complex, underlying dynamic (in this case, geometric exclusion rather than a Lorenz map) dictates the final, emergent spatial kinematics of the system.³¹

Critical Analysis and Open Areas for Investigation

While the initial data extracted from the single-frequency, singleton-group sector is highly compelling, the model's structural definitions inherently permit multi-dimensional expansions that remain entirely uncharted. To robustly validate this packing model as a comprehensive cosmological analogue, several critical theoretical gaps must be addressed.

1. Extrapolation to Absolute Saturation via Voxel-Subdivision

The most pressing computational vulnerability in the manuscript is the reliance on a stochastic acceptance-rate cutoff (10^{-6} , or one successful placement per million attempts) to declare the system "jammed".¹ While the local slopes of the scaling exponent remain remarkably flat, indicating the system is deep within the asymptotic regime, literal geometric saturation is typically only approached logarithmically.¹

To rigorously defend the precise value of $D = 2.33$, the data must be mathematically extrapolated to infinite time, or absolute saturation. The established RSA literature provides two distinct pathways to achieve this. The first is an asymptotic approximation using Feder's law, where the change in coverage over logarithmic time intervals is plotted to extract the true limit

$\theta(\infty)$.²²

However, a far superior method exists in the deployment of voxel-subdivision or exclusion-assisted packing algorithms.³⁵ In these advanced methodologies, the available parameter space is explicitly discretized into a voxel grid. When a worldline is accepted, all voxels in the parameter space that would result in an overlapping trajectory are geometrically flagged and deleted using intersection theorems (such as the Separating Axis Theorem for polygons).³⁵ The grid is recursively subdivided until the list of available voxels is empty, mathematically proving that no further insertions are possible. Implementing a

voxel-subdivision algorithm for the Bentley model would transition the N_{sat} measurement from a stochastic approximation into an exact, certifiable topological constant.³⁸

2. Multi-Frequency Worldlines and the Rapidity Partitioning

The current data is strictly limited to worldlines possessing a single non-trivial frequency per axis, enforcing $|ab| = 1$.¹ However, the foundational equation of the model defines the

rapidity budget as a summation over infinite potential frequencies: $\sum |ka_k| = 1$.¹ Evaluating the multi-frequency sector is critical because it will determine whether the

sub-dimensional exponent D and the $w = 0.070$ equation of state are universal thermodynamic properties of the rapidity constraint, or merely artifacts of the single-frequency truncation. Multi-frequency paths allow for highly localized, irregular "wiggles." This increased trajectory flexibility might allow multi-frequency worldlines to navigate through the dense thicket of an RSA jam far more efficiently than rigid, single-frequency sines.

Determining if the relaxation of the single-frequency rule pushes the capacity exponent D/d closer to unity, or leaves it invariant at ≈ 0.78 , is the most consequential next step for the kinematic theory.¹

3. Unique Groups, Subpaths, and Percolative Network Topology

The Bentley framework introduces the concept of "unique groups," defined as sets of paths that intersect members of their own group but remain strictly isolated from all other groups.¹

The current manuscript exclusively analyzes singleton groups (a hard-sphere analogue).

The intended secondary construction phase involves introducing "subpaths"—new trajectories that intentionally intersect an existing group member to merge their identity.¹ This transforms the model from a simple exclusion packing problem into an intricate network growth and percolation model. The physical interpretation of a unique group existing as a singular identity fragmented across multiple spatial locations introduces a mechanism for non-local correlation.¹ Investigating the phase transition threshold at which these multi-path groups percolate across the comoving torus will map the model's combinatorics onto established continuum percolation theory, providing profound insights into the connectivity of the packed universe.³⁹

4. Thermodynamic Evolution over Conformal Time ($w(z)$)

The calculation of the $w = 0.070$ equation of state is localized entirely at the turnaround slice ($z = \pi/2$), where the kinematic equations simplify exclusively to the phase parameters.¹ For the model to function as a viable cosmological analog, it must yield a coherent, unbroken evolutionary history from the initial bang ($z = 0$) to the final crunch ($z = \pi$).

Tracking the effective kinetic pressure and equation of state $w(z)$ continuously across the entire conformal loop is mandatory. The behavior of the packed worldlines as they approach the singularities is of particular interest. As $z \rightarrow 0$, the physical scale factor $a(z) \rightarrow 0$, forcing all worldlines to interact violently in localized physical space despite remaining perfectly uniformly distributed in the decoupled comoving space. Monitoring the thermodynamic response to this scale-factor collapse will test the absolute limits of the model's physical analogies and determine whether it exhibits epoch transitions (e.g., shifting from radiation-dominated to matter-dominated states).

5. Mathematical Validation of the Strange Attractor Hypothesis

To elevate the "strange attractor" hypothesis from an intuitive analogy to a proven mathematical reality, the authors must rigorously quantify the topology of the accepted parameter space.

If the iterative RSA algorithm functions as a genuine attractor, the accepted parameters (b, f, a_1) will not form a uniform distribution, but rather a complex fractal set embedded within the initial proposal measure. To prove this, the authors should extract the generalized dimension spectrum (including the capacity, information, and correlation dimensions) of the accepted parameter space itself, independent of the physical spatial slices.¹⁷ Furthermore, measuring the Lyapunov exponents of the insertion-rejection algorithm—specifically tracking how rapidly two adjacent, nearly identical random seeds diverge in their ultimate jammed compositions—would formally classify the system within the rigorous boundaries of nonlinear dynamical theory.¹⁴

Recommendations for Additional Experiments

To definitively address the open theoretical gaps and structurally reinforce the manuscript's profound initial findings, the following suite of specific, high-resolution experiments is highly recommended:

1. **Asymptotic Feder Scaling Calibration:** Execute a series of long-duration RSA runs on

small, highly tractable grid resolutions (e.g., $T = 20, 40, 60$), precisely tracking the

coverage fraction against the exact number of insertion attempts t . Plot

$\log[\theta(2t) - \theta(t)]$ versus $\log t$ to verify if the asymptotic approach strictly follows a

- power-law t^{-1/d_f} . Extract the exact exponent d_f to explicitly determine the effective degrees of freedom permitted by the global temporal exclusion constraint.²²
2. **Exact Saturation via Voxel-Subdivision:** Develop and implement a phase-space voxel-subdivision tree to mathematically certify absolute saturation for a low-resolution ($T = 20$) 2+1D system.³⁵ Compare this algorithmically proven saturation count against the 10^{-6} stochastic cutoff utilized in the current methodology to formally calibrate the error bars on the D exponent.
 3. **Multi-Frequency Kinematic Partitioning Tests:** Run a comparative RSA campaign allowing worldlines to possess exactly two frequencies per axis (e.g., $k = 2$ and $k = 3$) with a shared, partitioned rapidity budget ($2|a_2| + 3|a_3| = 1$). Evaluate the resultant exponent D , the parity composition, and the turnaround equation of state against the baseline single-frequency results to determine the universality of the kinematic constraints.
 4. **Full-Loop Thermodynamic Integration:** Utilizing a fully saturated $T = 100$ configuration, compute the mean squared peculiar velocity $\langle v^2 \rangle$ and the resulting kinetic pressure at every discrete timestep z_i . Plot $w(z)$ continuously from $z = 0$ to $z = \pi$ to trace the full thermodynamic epoch evolution of the model.
 5. **Parameter-Space Dimension Spectrum Analysis:** Apply the Grassberger-Procaccia algorithm directly to the final accepted values of the (b, f, a_1) parameter space rather than the physical X coordinates.¹⁷ Confirm whether the correlation dimension of the accepted parameters forms a fractal subset, thereby mathematically validating the strange attractor hypothesis.

Conclusion

The Bentley packing model represents a highly innovative, purely kinematic approach to theoretical cosmology. By substituting continuous quantum background fields with discrete, globally constrained, non-intersecting worldlines, the framework forces a re-evaluation of how fundamental physical properties emerge. The discoveries that this pure geometric exclusion

mechanism autonomously yields a sub-dimensional dynamical capacity ($D = 2.33$), exact spatial homogeneity without fine-tuning, spontaneous parity breaking, and an emergent matter-dominated turnaround state are phenomenologically profound.

The model successfully bridges deeply disparate disciplines, synthesizing the irreversible kinetics of Random Sequential Adsorption with the toroidal topological frameworks of String

Gas Cosmology and the internal state-space dynamics of attractor-driven active matter. By transitioning the computational methodology away from stochastic acceptance approximations toward rigorous algorithmic saturation proofs, mapping the continuous thermodynamic history $w(z)$, and investigating the percolative network topologies of multi-path unique groups, this model has the potential to evolve into a formidable theoretical instrument. It provides an entirely unique, path-centric lens through which to study how global temporal constraints can spontaneously generate structural and thermodynamic order without reliance on semiclassical quantum mechanics or explicit physical interaction potentials.

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